

Multi-Candidate Multimodal Advertisement Generation with Learning-Based Performance Prediction and Intelligent Ranking

Mr. Alexander Aruldoss

*Faculty, Department of Information Science and Engineering
Alva's Institute of Engineering and Technology, Mijar
Mangalore, Karnataka, India
alexander.aruldoss@aiet.org.in*

Disha

*Student, Department of Information Science and Engineering
Alva's Institute of Engineering and Technology, Mijar
Mangalore, Karnataka, India
dishagshetty6@gmail.com*

Sanket

*Student, Department of Information Science and Engineering
Alva's Institute of Engineering and Technology, Mijar
Mangalore, Karnataka, India
biradarsanket380@gmail.com*

Shankar C Akashkore

*Student, Department of Information Science and Engineering
Alva's Institute of Engineering and Technology, Mijar
Mangalore, Karnataka, India
shankarakashkore32@gmail.com*

Sujal Bandekar

*Student, Department of Information Science and Engineering
Alva's Institute of Engineering and Technology, Mijar
Mangalore, Karnataka, India
sujalbandekar786@gmail.com*

Abstract—Generating advertising creative with diffusion models is cheap at the still-image stage and expensive at the video stage. A single still costs roughly one-seventeenth of a single eight-to-ten-second clip, so a job that animates every candidate spends most of its budget on the last step. That shifts the useful question from *how good is one generated image to which of several candidates is worth animating*, a ranking problem that has to be answered one stage before any evidence for it exists. We present a multi-candidate pipeline built around that problem. Creative diversity is constructed geometrically instead of requested in words: a Latin-hypercube draw over four categorical design axes yields n candidate design points whose minimum pairwise Hamming distance is provably maximal, so candidates differ by construction and the construction can be ablated. Each design point is compiled into a structured prompt and rendered by a latent diffusion backbone conditioned on CLIP-family text embeddings and on reference images that carry model and product identity. Surviving frames pass a quality gate of five pixel-level measurements, whose thresholds are calibrated against a labelled render sweep instead of being picked by hand. Each candidate is then represented by features from frozen transformer and convolutional encoders, scored by a regression head trained under a Bradley–Terry objective, and ranked by weighted composite score. Only the promoted subset is animated and ranked again. We describe the architecture, survey the diffusion, vision-language and preference-learning literature it builds on, and state the implementation status as it stands. The sampler, brief compiler, generation path, quality gate, feature extraction, ranking, delivery layer and evaluation harness are implemented and exercised by 666 automated tests. The learned prediction head still awaits the human preference corpus it must be trained on, so the deployed scorer is a documented heuristic baseline, labelled as such wherever it is displayed. We report the

calibration measurements that exist, identify a cross-validation leakage effect that inflates apparent accuracy by nineteen points if left unguarded, and set out the evaluation protocol that remains to be run.

Index Terms—multimodal generation, latent diffusion, denoising diffusion probabilistic models, CLIP, vision transformer, Bradley–Terry model, learning to rank, preference learning, computational advertising, design-space sampling

I. INTRODUCTION

Text-conditioned diffusion models have made a plausible advertising still routine to produce, while a short advertising film is still expensive. That gap is the premise of this work. In the pipeline described below, generating one candidate still takes roughly two orders of magnitude less wall-clock time, and costs one order of magnitude less in compute, than generating an eight-to-ten-second clip from it. A job that generates five stills and animates all five therefore spends close to nine tenths of its budget on animation, spread evenly over candidates that are not equally good.

This puts the important decision at the boundary between the two stages: given several candidate stills, which deserve to become video? The text-to-image literature mostly asks a different question. It evaluates *single-sample fidelity* (does this image match this prompt, is it preferred, is it free of artefacts) over independently sampled images [1], [3], [8]. Advertising creative needs a *set* of candidates that covers the creative space, since a choice between five near-duplicates is no choice,

and a *preference order* over that set that is produced before the expensive stage instead of after it.

Both requirements are harder than they look. Diversity requested in natural language (“give me five different concepts”) produces variation that cannot be measured, reproduced under a fixed seed, or ablated. A preference order predicted at the still stage is also a claim about the video stage, so showing that it is worth anything takes paired observations at both stages and a protocol that does not compare the system with itself.

We describe a system built around these two requirements and report its current state, including what has not yet been done. The contributions are:

- 1) **Geometric design-space sampling.** A Latin-hypercube construction over four categorical axes (camera angle, lighting, composition, motion intent) guarantees the maximal minimum pairwise Hamming distance between candidates, so diversity is a controlled quantity instead of a hoped-for property (Section III-A).
- 2) **A calibrated pre-ranking quality gate.** Five pixel-level checks, with thresholds set on a labelled sweep of 150 lighting $\times$ composition $\times$ angle combinations, and a product-scale ceiling calibrated on real generations, reject structurally broken frames before any ranking effort goes into them (Section III-D).
- 3) **A frozen-encoder representation with an ablatable feature taxonomy.** Nine disjoint feature groups (photometric, compositional, salience, palette, contextual, embedding, aesthetic, identity and motion) are defined so that a group can be removed *exactly* and account for exactly one row of an ablation table (Section III-E).
- 4) **A two-stage ranking protocol with the confound stated.** The image-stage ranking is committed to the record before any video exists, and a test enforces that ordering. Model-to-model agreement between stages is reported as a diagnostic, kept apart from the human-preference claim it could otherwise be mistaken for (Sections III-G and IV-D).

Section II surveys the three literatures the work draws on. Section III describes the architecture module by module. Section IV reports implementation status, the calibration measurements so far, and the evaluation protocol still to be run. Section V concludes.

II. LITERATURE SURVEY

The relevant work falls into three strands: (i) conditional diffusion models for image and video synthesis, (ii) frozen vision-language and vision encoders used as general-purpose representations, and (iii) preference learning and ranking from pairwise human judgements. Table I maps representative work in each strand to the module of our architecture that uses it.

A. Conditional Diffusion Synthesis

Denosing diffusion probabilistic models [1] cast generation as the reversal of a fixed Gaussian corruption process and introduced the noise-prediction objective that our generation

stage inherits. Song et al. [2] gave a deterministic, non-Markovian sampler that needs far fewer function evaluations, which made diffusion affordable inside an interactive pipeline. Rombach et al. [3] moved diffusion into the latent space of a pretrained autoencoder, cutting the cost of high-resolution synthesis by roughly an order of magnitude, and introduced the cross-attention mechanism that the brief compiler writes into. Classifier-free guidance [4] trades diversity against conditioning fidelity at inference. Because our sampler produces diversity upstream, strong guidance can be applied downstream without collapsing the candidate set. Peebles and Xie [5] replaced the convolutional U-Net with a transformer, and Liu et al. [6] and Esser et al. [7] reformulated the corruption path as a rectified flow. Our generation stage targets that class of backbone, not one specific checkpoint. Ho et al. [9] extended diffusion to video, and Blattmann et al. [10] showed image-conditioned video diffusion, which is the operation our animation stage applies to a promoted still. None of this work supplies a selection criterion. Each paper evaluates samples, and none asks which of several should be promoted to a more expensive downstream process. That gap is what the prediction and ranking stages address.

B. Frozen Encoders as Representations

CLIP [11] showed that contrastive image-text pretraining at scale yields an embedding that transfers well beyond its training task, and it supplied the text encoder that conditions most later diffusion models. Zhai et al. [12] replaced CLIP’s softmax contrastive loss with a pairwise sigmoid loss, which improves quality at small batch sizes. SigLIP is the CLIP-family encoder we use for the semantic feature block. DINOv2 [13] provides self-supervised visual features that keep fine-grained appearance detail, where CLIP-family embeddings are weakest, and we use it for product-identity similarity. ArcFace [14] supplies the additive angular margin embedding behind the face-identity check, which guards against drift from the model’s likeness. The LAION aesthetic predictor [15] contributes a single scalar, which we keep as its own ablation group instead of folding it into the general embedding block. The annotation corpus holds hundreds of comparisons, not millions, so all four encoders are frozen and only a small head is fitted. Section III-F quantifies that choice.

C. Preference Learning and Ranking

The Bradley–Terry model [16] gives each item a scalar latent strength, and the probability that one item beats another is a logistic function of the difference in strengths. Hunter’s minorization-maximization iteration [17] fits it and converges monotonically without a learning rate. Burges et al. [18] showed that the same pairwise logistic objective can sit on top of a parametric scorer, which turns ranking into supervised learning on comparisons. Our prediction head adopts this formulation. ImageReward [19], PickScore [20] and Human Preference Score [21] each fit a preference model on a large corpus of human choices between generated images, and each shows that learned preference predicts human judgement better

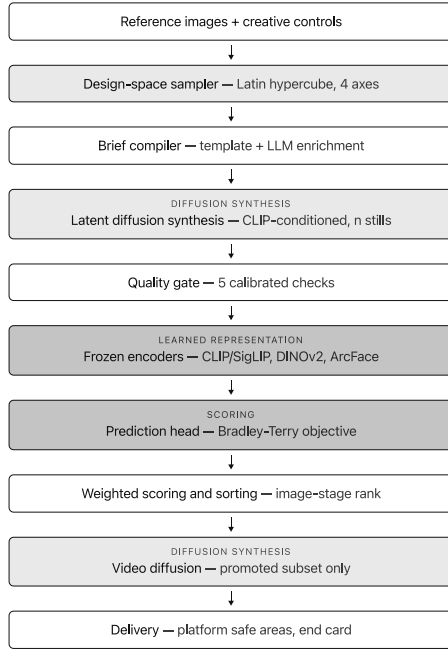


Fig. 1. Pipeline architecture. Shaded stages are diffusion synthesis (light) and learned representation and scoring (dark). The dashed boundary marks where the ranking decides how the remaining budget is spent; only promoted candidates cross it.

than fidelity metrics such as FID or CLIP-score. These are the closest antecedents to our work, but the target differs. All three rank images to *evaluate a generator*. Our system ranks candidates within a job to *allocate a downstream budget*, and its target variable is one modality removed from the input. The class-imbalance and annotator-agreement problems those papers document also apply to us, with greater force, because our corpus is smaller by orders of magnitude.

III. IMPLEMENTATION METHODOLOGY

The system is organised as nine sequential stages joined by an append-only job record (Fig. 1). Two properties of that record matter for the architecture. First, each stage writes its output before the next stage reads it, so the image-stage ranking is physically committed before any video exists, and an automated test enforces that ordering. Second, every measurement carries an explicit *implemented* flag, so a check that has not been built reports that it is missing instead of defaulting to a pass.

A. Design-Space Sampling

We model the creative space as $A = 4$ categorical axes (camera angle, lighting, composition and motion intent) with cardinalities $|L_a|$ for $a \in \{1, \dots, A\}$. To produce n candidates, the sampler draws a subset of n *distinct* levels independently on each axis and pairs them positionally:

$$d_i = (\ell_{\pi_1(i)}^{(1)}, \ell_{\pi_2(i)}^{(2)}, \dots, \ell_{\pi_A(i)}^{(A)}), \quad i = 1, \dots, n, \quad (1)$$

where π_a is an independent permutation of the n chosen levels on axis a . Each axis contributes n distinct levels and

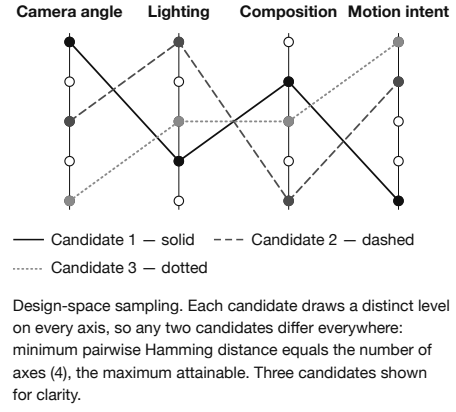


Fig. 2. Three example candidates under the sampler of (1).

each candidate takes a different one, so any two candidates differ on *every* axis, which gives

$$\min_{i \neq j} \text{Hamming}(d_i, d_j) = A \quad \text{whenever } n \leq \min_a |L_a|. \quad (2)$$

This is the maximum attainable value, so there is nothing left to optimise; the construction is optimal by definition, not by search. What (2) buys in practice is that diversity becomes an *independent variable*. We can ablate the sampler, either by replacing it with an unconstrained random draw or by repeating a single design point, and measure the effect on downstream ranking quality. That is impossible when diversity is delegated to a language model’s sense of variety.

Two mechanisms modulate the draw without breaking the guarantee. A *mood* parameter reweights which levels are preferred on each axis; if the preferred pool cannot supply n distinct values, the draw widens to the full axis instead of duplicating a level. A user-locked camera angle collapses that axis to a constant and lowers the bound in (2) to $A - 1 = 3$. The sampler reports the reduced distance instead of hiding it.

The sampler also applies one structural exclusion. The `negative_space_top` composition places the subject low in the frame to leave room for overlaid text. On vertical placements that puts the subject beneath platform chrome, which covers the bottom 18 to 22% of the frame. The measured safe-area saliency share for this composition is 0.44 to 0.67, against 0.68 to 0.91 for every other composition, so it fails the gate structurally and not just occasionally. We exclude it in the sampler because otherwise we would pay to generate a candidate that is guaranteed to be rejected.

B. Brief Compilation

A template compiles each design point into a generation prompt and writes the structural constraints out explicitly: the positional roles of the reference images, the camera angle and lighting phrases, the composition, the brand palette and the safe-area requirement. Naming the references by position, meaning which input carries the model’s likeness and which carries the product, is the most effective intervention against

TABLE I
SUMMARY OF REVIEWED LITERATURE AND RELEVANCE TO THE PRESENT SYSTEM

Author(s) / Year	Focus	Contribution	Relevance to this system
Ho et al. (2020) [1]	Diffusion generation	Introduced DDPM; generation as reversal of a fixed Gaussian corruption process, trained by noise prediction	Foundational basis for the generation stages
Song et al. (2021) [2]	Accelerated sampling	Deterministic non-Markovian sampler; order-of-magnitude fewer denoising steps	Keeps per-candidate generation cheap enough for a multi-candidate job
Rombach et al. (2022) [3]	Latent diffusion	Diffusion in autoencoder latent space with cross-attention text conditioning	Backbone architecture and conditioning path used here
Ho & Salimans (2022) [4]	Conditioning control	Classifier-free guidance; trades diversity for prompt fidelity at inference	High guidance is safe since diversity comes from the sampler
Peebles & Xie (2023) [5] Liu et al. (2023); Esser et al. (2024) [6], [7]	Denoiser architecture Flow matching	Transformer denoiser replacing the convolutional U-Net Rectified-flow straight-line corruption paths; improved sample efficiency	Backbone class the generation stage targets Backbone-agnostic interface; current backbones combine this with DiT
Ho et al. (2022); Blattmann et al. (2023) [9], [10] Radford et al. (2021) [11] Zhai et al. (2023) [12] Oquab et al. (2023) [13] Deng et al. (2019) [14] Schuhmann et al. (2022) [15] Bradley & Terry (1952) [16] Hunter (2004) [17]	Video diffusion Vision-language pretraining Vision-language pretraining Self-supervised vision Face representation Aesthetic scoring Paired comparison Estimation	Contrastive image-text alignment; transferable joint embedding space Pairwise sigmoid loss removing batch-global normalisation Fine-grained visual features without language supervision Additive angular margin loss for discriminative face embeddings Linear aesthetic head over CLIP embeddings Latent-strength model for pairwise outcomes MM iteration for Bradley-Terry; monotone convergence, no learning rate	The operation applied to a promoted still at animation Text encoder conditioning generation and the semantic feature block CLIP-family encoder used for the semantic embedding feature group Product-identity similarity against the reference photograph Face-identity check guarding against model-likeness drift Scalar aesthetic feature, isolated as its own ablation group Objective for the prediction head and the ranking fit Fitting procedure used for latent strengths
Burges et al. (2005) [18] Xu et al. (2023); Kinstain et al. (2023); Wu et al. (2023) [19]–[21]	Learning to rank Generative preference	Pairwise logistic objective over a parametric scorer Learned human-preference models outperform FID/CLIP-score	Formulation of the trainable prediction head Closest antecedents; this system ranks within a job, not across generators

identity drift. It is also exactly the material that a language model asked to be creative will paraphrase away.

For that reason a language model enriches *only* the free-form scene description and a one-line concept, and the structural section never passes through it. If the language model is unavailable or returns malformed output, the template result stands alone and the job proceeds. This fallback is not an exceptional path: every offline test run takes it, so it is exercised continuously.

C. Conditional Image Synthesis

Generation follows the standard latent diffusion formulation [1], [3]. An encoder maps an image to a latent z_0 ; a forward process adds Gaussian noise over T steps according to a variance schedule $\{\beta_t\}$,

$$q(z_t | z_{t-1}) = \mathcal{N}(z_t; \sqrt{1 - \beta_t} z_{t-1}, \beta_t \mathbf{I}), \quad (3)$$

which admits the closed form $z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$ with $\bar{\alpha}_t = \prod_{s \leq t} (1 - \beta_s)$ and $\epsilon \sim \mathcal{N}(0, \mathbf{I})$. A denoiser ϵ_θ is trained to invert it under the conditioning signal c :

$$\mathcal{L} = \mathbb{E}_{z_0, \epsilon, t} \left[\left\| \epsilon - \epsilon_\theta(z_t, t, c) \right\|_2^2 \right]. \quad (4)$$

The conditioning signal c has two components. The first is text: the compiled brief, embedded by a CLIP-family transformer encoder [11], [12] and injected through cross-attention. The second is the pair of reference images, which carry model and product appearance through the backbone’s image-conditioning path. This second component makes the task editing and composition instead of free synthesis, and it is why identity preservation can be measured at all. At inference, classifier-free guidance [4] combines conditional and unconditional predictions,

$$\tilde{\epsilon}_\theta(z_t, t, c) = \epsilon_\theta(z_t, t, \emptyset) + w(\epsilon_\theta(z_t, t, c) - \epsilon_\theta(z_t, t, \emptyset)), \quad (5)$$

with guidance weight w . Sampling uses a deterministic solver [2] and each candidate carries a fixed seed, so a job is reproducible: the same design point, brief and seed yield the same latent trajectory. The research design requires this, because the sampler ablation of (1) can only be interpreted if the downstream generation step is held constant.

TABLE II
QUALITY-GATE CHECKS, MEASURED ON-BRAND RANGES, AND THRESHOLDS

Check	On-brand range	Threshold	Catches
Palette adherence	0.912–0.972 (off-brand 0.000)	≥ 0.456	Brand-colour violation
Safe-area saliency	0.680–0.914	≥ 0.55	Subject under platform chrome
Focal clarity	0.578–0.728	≥ 0.25	Smearred attention, no subject
Luminance	0.531–0.643	$[0.06, 0.96]$	Black or blown frames
Contrast	0.180–0.248	≥ 0.03	Flat or empty frames
Product area share	see Sec. III-D	≤ 0.16	Product larger than the person

D. Quality Gate

Five pixel-level measurements screen candidates before any ranking effort is spent on them. We fitted the thresholds with a calibration sweep over 150 lighting $\times$ composition $\times$ angle combinations, rendered with matched and deliberately mismatched brand palettes. Each threshold sits *below the observed minimum* of the legitimate class instead of at its median, because a filter that rejects a third of usable candidates costs more than it saves. Table II lists the measured ranges and the thresholds chosen.

The product-area check is the only one with a ceiling instead of a floor, and the only threshold calibrated on real generations instead of the synthetic sweep. We added it because early live jobs rendered the product far larger than the person holding it, a failure that none of the existing checks expressed. Focal clarity over the same eleven frames spans 0.31 to 0.50, with correct and oversized frames interleaved, so it separates nothing. Product area share does:

visibly correct scale	0.011–0.094	($n = 5$)
unlabelled	0.084–0.113	($n = 3$)
visibly oversized	0.172–0.294	($n = 3$)

The classes separate with a gap of 0.059. The threshold sits near the top of that gap, for the same asymmetry as above: a false reject costs a paid regeneration, while a miss only lets one more frame into ranking. Eleven frames make a calibration set and not a validation set, and the code flags this as the first threshold due for revision as the corpus grows.

Two further checks compare face and product identity against the reference images. They compute a real cosine similarity, using ArcFace and DINOv2 respectively, whenever an embedding archive is available, but they still report

TABLE III
FEATURE GROUPS AND THEIR PROVENANCE

Group	Content	Source
Photometric	Exposure, contrast, saturation	Pixels
Composition	Position and extent of the salient mass	Pixels
Salience	Saliency concentration, safe-area compliance	Pixels
Palette	Brand-palette adherence	Pixels
Context	The sampler’s design coordinate, vertical, platform	Job record
Embedding	Reduced CLIP/SigLIP and DINOv2 blocks	Frozen encoder
Aesthetic	LAION aesthetic score	Frozen encoder
Identity	DINOv2 product and ArcFace face similarity to references	Frozen encoder
Motion	Motion energy, temporal consistency, hook strength	Video only

implemented = false unconditionally. No calibration sweep has measured where same- and different-identity pairs separate for this generator, so any threshold we set now would be invented and not calibrated. The measured value is still surfaced and propagated as *pending*, so no interface can present it as a verified pass. The missing piece is not the computation, which is tested end to end, but the fact that nothing in the synchronous request path can run a GPU encoder inline.

E. Representation

Each surviving candidate is represented by a feature vector assembled from nine disjoint groups (Table III). The partition is the unit of ablation, so it has to be exact: a group must be removable without residue, because an ablation that leaves a correlated proxy behind reports a contribution that is not there.

Ablating the *context* group answers a research question directly. The group contains only the design coordinate the sampler chose, so removing it tests whether the creative decision carries information independent of how the render turned out.

The frozen encoders run as a separate offline stage that writes a compressed embedding archive, which the trainable head consumes. The head is implemented once and used by both the evaluation harness and the serving path, so the model reported in an ablation table is bit-identical to the model the system scores with. There is no train/serve reimplement gap that could make the reported numbers wrong.

This stage guards against silent zero-filling. An item without an embedding must not receive a zero vector, because after standardisation a zero vector *is* the feature-space mean and would look average instead of missing. Coverage is returned alongside the matrix, and uncovered items are dropped explicitly. Deterministic pseudo-embeddings also exist so the training path can be exercised without running the encoders. A synthetic flag propagates into the evaluation output, so any table built on them says so.

F. Performance Prediction

The prediction head is a parametric scorer s_θ over the feature vector $\phi(x)$, trained directly on pairwise comparisons under the Bradley–Terry likelihood [16], [18]. For a comparison in which candidate i was preferred to candidate j ,

$$\Pr(i \succ j) = \sigma(s_\theta(\phi(x_i)) - s_\theta(\phi(x_j))), \quad \sigma(u) = \frac{1}{1 + e^{-u}}, \quad (6)$$

and the head is fitted by minimising the regularised negative log-likelihood over the observed comparison set $\mathcal{C}$:

$$\mathcal{L}(\theta) = - \sum_{(i,j) \in \mathcal{C}} w_{ij} \log \sigma(s_\theta(\phi(x_i)) - s_\theta(\phi(x_j))) + \lambda \|\theta\|_2^2. \quad (7)$$

Ties are kept and enter as two half-weight observations in opposing directions. Discarding them would remove the observations where candidates are close, which is where a ranker’s behaviour matters most.

The number of labels, not the encoders, sets model capacity. The default head is linear. With approximately 1,200 training comparisons, a defensible parameter budget is on the order of 80, while a single eight-unit hidden layer over thirty inputs already needs roughly 250. We implemented a hidden-layer variant so the report can *show* what exceeding that budget does. It gets one table row and is explicitly not the headline model.

When latent strengths are estimated from comparisons alone, as when ranking without features, the Bradley–Terry model is fitted with Hunter’s minorization-maximization iteration [17]. It converges monotonically and needs no learning rate, autograd or GPU. Under plain MM, an item that wins every comparison it appears in has unbounded strength. A phantom-opponent prior, meaning a few games credited against a fixed-strength opponent with half won and half lost, keeps the estimates finite and pulls sparsely observed items toward the centre.

We also implemented an alternative two-step formulation that fits strengths first and then regresses features onto them. The theoretical objection is that squared error over-weights the least-observed, most-extreme strengths. That did not appear in our simulations: the two-step fit stayed within a point of the direct fit over simulated corpora of 25 to 200 sets. The pairwise objective remains primary anyway, because it needs no intermediate fit and treats ties as observations.

G. Ranking and Promotion

Candidate scores are combined into a composite by weighted scoring and sorting. For candidate i with component scores c_{ik} and weights w_k summing to one,

$$S_i = \sum_k w_k c_{ik}, \quad \text{rank}(i) = |\{j : S_j > S_i\}| + 1, \quad (8)$$

with ties broken deterministically by identifier. The top- m candidates by S_i are promoted. The rest stay in the record with their scores and gate measurements, so a job documents what it declined to animate and why. The image-stage ranking is written to the record *before* any video generation is dispatched. The research claim depends on this ordering, and a test enforces it instead of relying on developer discipline.

After the promoted candidates are animated, the finished videos are ranked again over a feature set extended by the motion group. The quantity of interest is the agreement between the two orderings, measured by Spearman’s rank correlation

$$\rho = 1 - \frac{6 \sum_{i=1}^m d_i^2}{m(m^2 - 1)}, \quad d_i = r_i^{\text{img}} - r_i^{\text{vid}}, \quad (9)$$

together with pairwise agreement accuracy and bootstrap intervals, since a point estimate over a few hundred comparisons is not a result. Each job contributes one paired observation, so the evaluation dataset grows as the system is used. We report the cost consequence as a counterfactual: had only the top-ranked still been animated, how often would the human-preferred video have been retained? This quantifies the saving without the system having had to make the cut. Section IV-D describes the confound that limits how (9) can be interpreted at present.

H. Video Synthesis and Delivery

An image-conditioned video diffusion model [9], [10] animates each promoted still into a clip of eight to ten seconds, with the motion intent from (1) supplying camera and subject motion. A composited end card carrying the brand mark and web address is appended within the clip’s declared duration at no additional cost. Delivery renders each clip for seven placement targets across four aspect ratios and applies the same platform safe-area inset that the gate’s saliency check was calibrated against.

I. Technology Stack

The orchestration API is Python/FastAPI on top of an SQLAlchemy-backed relational store that holds the append-only job records, the spend ledger and the annotation corpus. Pipeline stages, feature extraction, ranking and evaluation are written in NumPy with no deep-learning framework dependency, which keeps the trained head identical across evaluation and serving. The frozen encoders run in a separate GPU environment and communicate through a compressed embedding archive. The operator interface is TypeScript/Next.js. Its client-side types are generated from the Python schema by a drift-checked generator, so the two halves cannot disagree about a job’s shape.

IV. RESULT AND ANALYSIS

The system is a mid-development prototype and not a deployed product. This section reports its status and does not present evaluation figures that have not yet been produced.

A. Implementation Status

The design-space sampler, brief compiler, conditional generation path, quality gate, feature extraction, ranking, video animation, multi-platform delivery, spend ledger and evaluation harness are implemented and exercised end to end. The 666 automated tests include ones that enforce the research-critical invariants: the image-stage ranking is committed before video generation, pending gate checks cannot present as passing measurements, and a replayed job collapses onto the job it replays instead of counting as a second observation.

The learned prediction head is implemented, together with its training script, cross-validation protocol, baseline comparisons and per-group ablation. It has not been trained on real human preference labels, because the annotation corpus does not exist yet. **The scorer currently deployed is therefore a**

documented, hand-weighted heuristic baseline, and every interface that displays one of its scores flags it as a placeholder. Three conditions mark a score as a placeholder: a model trained on stand-in embeddings, a model whose held-out accuracy was never measured, and a model measured at chance. The face- and product-identity gate checks now compute a real cosine similarity against an embedding archive when one is supplied (Section III-D), and tests exercise this against synthetic archives. On a live job today no archive is supplied, so both checks still report as pending, and their thresholds await the same kind of calibration sweep that every other check in Table II required. No trained-model accuracy figure exists, and we do not claim one.

B. Experimental Setup

The measurements below were not produced by a benchmark harness against a trained model, since none exists yet. This subsection therefore says what was run to produce every number here and in Section IV-D. Three independent calibration protocols underlie them. They share no data with one another, and none touches the still-untrained prediction head. The gate’s five pixel-level thresholds (Table II) were fitted against 150 lighting $\times$ composition $\times$ angle combinations rendered with matched and deliberately mismatched brand palettes. The product-area ceiling was calibrated separately, against eleven hand-labelled frames from two real, paid jobs, because the synthetic sweep does not reproduce the failure it targets. The cross-validation leakage measurement (Section IV-D) used 60 synthetic generation sets whose features were constructed to be uninformative. True pairwise accuracy on them is therefore chance, and any deviation from chance comes from the split and not from signal in the features. This isolates the leakage mechanism from the separate, still-open question of whether real features carry signal.

C. Preliminary Observations

The measurements that exist are calibration measurements, reported in Section III and summarised here. The gate’s five primary thresholds were fitted against a 150-combination render sweep in which the on-brand and off-brand palette classes separate with a gap of 0.912 (Table II, Fig. 3). By construction, each threshold sits below the legitimate class’s observed minimum. The product-area ceiling was calibrated against eleven frames from real generations, where the correctly scaled and oversized classes separate with a gap of 0.059 while focal clarity fails to separate them at all (Fig. 4). The compositional exclusion in the sampler was measured in the same way: 0.44 to 0.67 safe-area saliency share, against 0.68 to 0.91 for the alternatives.

On a representative completed job, the cost split matched the prediction in Section I. The five candidate stills were generated and gated for approximately 15% of the job total, and the two promoted candidates took the remaining 85% at animation (Fig. 5). This is a spend measurement and not a quality one, and it confirms that the budget is concentrated where the ranking decision sits. The unit prices are fixed provider prices,

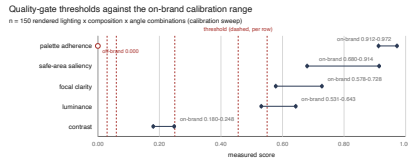


Fig. 3. The five quality-gate thresholds of Table II against their calibration ranges. Every threshold sits below its class’s observed minimum. The palette check’s off-brand point is shown hollow for contrast.

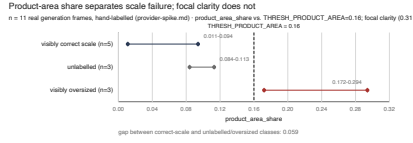


Fig. 4. Product-area share separates the three labelled classes with a 0.059 gap. Focal clarity, measured over the same eleven frames, spans 0.31 to 0.50 with correct and oversized frames interleaved, so it cannot.

not estimates: a still costs \$0.04 and a ten-second clip \$0.70, a ratio of $17.5\times$ that matches the “one-seventeenth” figure in Section I. The default job of five stills and two promoted videos therefore costs \$1.60, against a \$35.00 total budget tracked by the same ledger. Animating every candidate would cost \$3.70 for the same five stills, which is the cost the ranking stage exists to avoid.

Figure 6 shows the actual candidates from one completed job (\$1.60, five stills, two promoted) instead of a schematic. It gives the image-stage order that the ranking committed to the record, each candidate’s heuristic-baseline placeholder score, and which two were promoted. The video stage reordered the promoted pair: the image stage predicted $B > C$, and the finished videos ranked $C > B$. This is a concrete instance of the confound that Section IV-D states in the abstract. An image-stage prediction is a claim about the video stage, and this job is one observation of that claim being tested. It has not yet been validated against human judgement.

D. Threats to Validity

We identified three methodological hazards during construction. Each can produce a result that looks plausible but is fictional.

Cross-validation leakage across generation sets. Comparisons from the same generation set share candidates, so a split randomised over *comparisons* puts near-duplicate observations on both sides of the fold boundary. On features that are uninformative by construction, so that true accuracy is chance, a comparison-level split reports 0.659 pairwise accuracy and a set-level split reports 0.465 (Fig. 7). The choice of split alone accounts for nineteen points of accuracy that are not real. We therefore always cross-validate over generation sets and never over comparisons.

Model-to-model agreement is not the research claim. The claim is that image-stage prediction anticipates how people prefer finished *videos*. Both stages are scored by the same codebase over overlapping feature groups, so some agreement is guaranteed by construction, and (9) computed today measures internal consistency and not predictive validity.



Fig. 5. Spend split for a representative job, with the unit prices behind it. Video is $17.5\times$ the price of a still, so promoting two of five instead of animating all five is the difference between \$1.60 and \$3.70.

We therefore report agreement as a diagnostic and report the human comparison as *absent* until video-stage judgements exist. Quoting the first as the second would be the most likely way for this project to overclaim.

Replay inflation. Deterministic replay of a stored job writes a fresh record with a new identifier and bit-identical candidates. If observations were counted by job identifier, a replay script run twenty times would report twenty sets in perfect agreement, with a zero-width confidence interval. That would be a fabricated headline result from code that looks correct. Sets are therefore keyed by a hash of their candidates’ content digests, so a replayed job collapses onto the job it replays. If two records share a set key but disagree in order, that is scorer nondeterminism, and it is surfaced as a defect and not averaged away.

E. Limitations and Planned Evaluation

The central limitation of this paper is that five results a reader would expect are not yet available. We name them here so that nobody has to infer them from their absence: a trained-model performance figure, a human preference dataset, baseline comparison results, per-group ablation results, and human video-preference validation. All five depend on the same missing input, the pairwise human preference corpus described in Section III-F, which does not exist yet. None of them is approximated, simulated or estimated from proxy data here, because that would produce the kind of plausible but fictional result that Section IV-D guards against. The deployed scorer is therefore a documented heuristic baseline throughout, and it is never presented as a trained model.

Once the corpus exists, we will run the following protocol: (i) collect pairwise preference judgements over image-stage candidates and, separately, over the finished videos, with enough overlap to estimate inter-annotator agreement; (ii) establish the accuracy ceiling that a perfect predictor could reach against single noisy judgements, and report accuracy as a fraction of it; (iii) evaluate the trained head against random, aesthetic-only and gate-score-only baselines as *paired* comparisons with bootstrap intervals; (iv) run the per-group ablation of Table III against the full model, paying particular attention to the context group; (v) report the image-to-video rank agreement of (9) against human judgements, together with the promotion counterfactual; and (vi) ablate the sampler of (1) against an unconstrained draw. A confidence interval wider than the gaps between models is itself the finding.

V. CONCLUSION

This paper presented a multi-candidate multimodal advertisement generation pipeline built around a cost asymmetry.

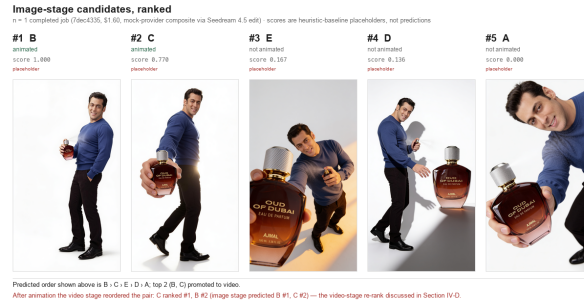


Fig. 6. Image-stage candidates from a real completed job (id 7dec4335, product “Perfume”, \$1.60), in the order the ranking committed to the record. Scores are the heuristic baseline described in Section IV-A, not a trained prediction. After animation, the video stage ranked C above B, reversing the image-stage order shown here for the promoted pair.

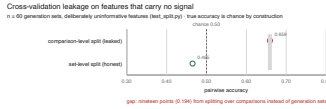


Fig. 7. The leakage effect on features built to carry no signal. Splitting over comparisons instead of generation sets produces nineteen points of accuracy that are not real.

Still-image synthesis is cheap and video synthesis is not, so the decision that matters is which candidates to animate, and it has to be made one stage before the evidence for it exists. The system addresses this with a Latin-hypercube design-space sampler whose diversity guarantee is provable and ablatable; a latent diffusion generation stage conditioned on CLIP-family text embeddings and identity-carrying reference images; a quality gate whose thresholds are calibrated on labelled render sweeps and not picked by hand; a frozen-encoder representation partitioned into exactly ablatable feature groups; and a Bradley–Terry-trained prediction head that feeds a weighted scoring and sorting stage, which commits its ranking to the record before any video exists.

As Section IV reports, the generation, gating, representation, ranking, animation, delivery and evaluation infrastructure is implemented and covered by 666 automated tests. The learned prediction head still awaits its human preference corpus, and the deployed scorer remains a documented heuristic baseline that is marked as a placeholder wherever it appears. We claim no trained-model performance figure. The immediate next steps are to collect the pairwise annotation corpus, run the evaluation protocol of Section IV, and calibrate the two pending identity checks. Longer term, the image-stage predictor should be retrained continuously on the paired observations that the system accumulates in normal operation. That is what lets the ranking claim be tested at increasing scale instead of just once.

REFERENCES

- [1] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 6840–6851.
- [2] J. Song, C. Meng, and S. Ermon, “Denoising diffusion implicit models,” in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2021.
- [3] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, “High-resolution image synthesis with latent diffusion models,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 10684–10695.
- [4] J. Ho and T. Salimans, “Classifier-free diffusion guidance,” *arXiv:2207.12598*, 2022.
- [5] W. Peebles and S. Xie, “Scalable diffusion models with transformers,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 4195–4205.
- [6] X. Liu, C. Gong, and Q. Liu, “Flow straight and fast: Learning to generate and transfer data with rectified flow,” in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2023.
- [7] P. Esser *et al.*, “Scaling rectified flow transformers for high-resolution image synthesis,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2024.
- [8] D. Podell *et al.*, “SDXL: Improving latent diffusion models for high-resolution image synthesis,” in *Proc. Int. Conf. Learn. Representations (ICLR)*, 2024.
- [9] J. Ho, T. Salimans, A. Gritsenko, W. Chan, M. Norouzi, and D. J. Fleet, “Video diffusion models,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2022.
- [10] A. Blattmann *et al.*, “Stable video diffusion: Scaling latent video diffusion models to large datasets,” *arXiv:2311.15127*, 2023.
- [11] A. Radford *et al.*, “Learning transferable visual models from natural language supervision,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2021, pp. 8748–8763.
- [12] X. Zhai, B. Mustafa, A. Kolesnikov, and L. Beyer, “Sigmoid loss for language image pre-training,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 11975–11986.
- [13] M. Oquab *et al.*, “DINOv2: Learning robust visual features without supervision,” *Trans. Mach. Learn. Res. (TMLR)*, 2024.
- [14] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, “ArcFace: Additive angular margin loss for deep face recognition,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 4690–4699.
- [15] C. Schuhmann *et al.*, “LAION-5B: An open large-scale dataset for training next generation image-text models,” in *Adv. Neural Inf. Process. Syst. (NeurIPS) Datasets and Benchmarks*, 2022.
- [16] R. A. Bradley and M. E. Terry, “Rank analysis of incomplete block designs: I. The method of paired comparisons,” *Biometrika*, vol. 39, no. 3/4, pp. 324–345, 1952.
- [17] D. R. Hunter, “MM algorithms for generalized Bradley–Terry models,” *Ann. Statist.*, vol. 32, no. 1, pp. 384–406, 2004.
- [18] C. Burges *et al.*, “Learning to rank using gradient descent,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2005, pp. 89–96.
- [19] J. Xu *et al.*, “ImageReward: Learning and evaluating human preferences for text-to-image generation,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2023.
- [20] Y. Kirstain, A. Polyak, U. Singer, S. Matiana, J. Penna, and O. Levy, “Pick-a-Pic: An open dataset of user preferences for text-to-image generation,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2023.
- [21] X. Wu *et al.*, “Human preference score: Better aligning text-to-image models with human preference,” in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023.